

1 Introduction

Question: What is Calculus?

Answer: Calculus is the Mathematics of change.

We may ask: How fast is a car moving at this instant? How much water has flowed into a tank up to this time? How does the temperature of a room change? How do engineers optimize the shape of a bridge or an aircraft wing? These are all questions involving continuous change. Whenever something changes continuously, calculus provides the mathematical language to study and analyze it.

Where is Calculus Used?

Calculus finds applications in almost every branch of science and engineering. Some examples include:

- Mechanical Engineering: velocity, acceleration, and the motion of machines.
- Electrical Engineering: current and voltage varying with time, and signal processing.
- Computer Science: optimization, machine learning, and neural networks.
- Civil Engineering: beam deflection and structural analysis.
- Chemical Engineering: reaction rates and diffusion.
- Data Science: optimization algorithms and error minimization.

These are only a few examples; calculus is a fundamental tool in many other disciplines as well.

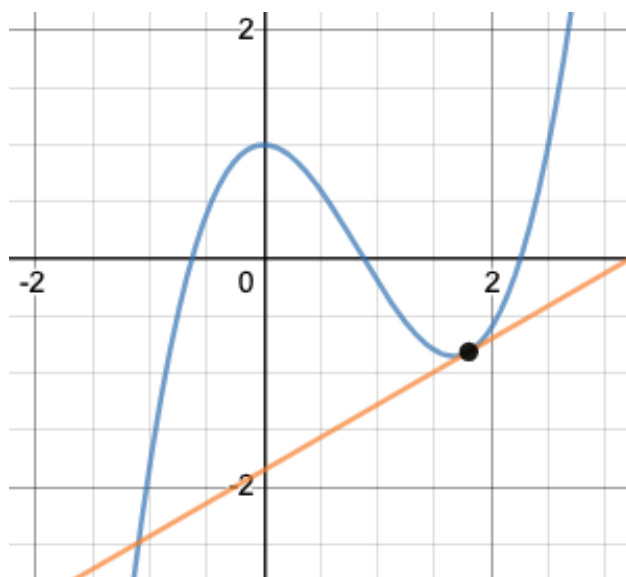
Calculus is a branch of mathematics concerned with the study of sequences, functions, limits, continuity, derivatives, integrals, and infinite series. It provides the mathematical language for understanding and analyzing continuous change.

Sir Isaac Newton (1642–1727), an English mathematician, physicist, and astronomer, and Gottfried Wilhelm Leibniz (1646–1716), a German mathematician, philosopher, and scientist, independently developed the foundations of calculus during the late seventeenth century.

1.1 Syllabus

The syllabus of the course is as follows. The course code is MA1110, and in this course, we will mainly discuss these three topics:

1. Sequence of real numbers: $\{a_1, a_2, \dots\}$.
2. Series of real numbers: $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$.
3. Differential calculus, where we will discuss continuity and differentiability.



1. Sequence of Real Numbers.

By a sequence of real numbers, we mean an enumerated infinite collection of real numbers indexed by the set of natural numbers. Here, we will study the notion of limits for a sequence. If a sequence has a limit, then we call such a sequence a convergent sequence. We will study various properties of convergent sequences, and we will see many examples. Next, we study monotone sequences, that is, either monotone increasing or monotone decreasing sequences. At the end of this section, we give a criterion called the Cauchy criterion, which gives a necessary and sufficient condition for a sequence to be convergent without having any knowledge of the limit.

2. Series of Real Numbers.

In the second section, we discuss a series of real numbers. By a series of real numbers, we roughly mean adding infinitely many real numbers. The study of series is a major part of calculus. Series are used in most areas of mathematics, even for studying finite structures such as combinatorics through generating functions. Infinite series are also widely used in other disciplines such as physics, computer science, statistics, and finance.

For a long time, the idea that such a potentially infinite summation could produce a finite result was considered paradoxical. This paradox was resolved using the concept of a limit during the 17th century. This infinite summation can have a finite value—in this case, the series is called a convergent series; otherwise, it is called a divergent series. We will study various tests to determine whether a series is convergent or divergent.

3. Differential Calculus.

In the last part of this course, we will study differential calculus, which is a subfield of calculus that studies the rates at which quantities change. It is one of the two traditional divisions of calculus, the other being integral calculus, which is the study of the area beneath a curve. The primary objects of study in differential calculus are the derivative of a function and related notions. The derivative of a function at a chosen point describes the rate of change of the function near that point. Geometrically, this is the tangent at this point of this curve, and the derivative is given by the slope of this tangent line.

Differential calculus and integral calculus are connected by the Fundamental Theorem of Calculus, which states that differentiation is the reverse process to integration. Differentiation has applications to nearly all quantitative disciplines. For example, in physics, the derivative of the momentum of a body with respect to time equals the force applied to the body. Rearranging this statement leads to the famous equation $F = ma$ associated with Newton's second law of motion. In operations research, derivatives determine the most efficient ways to transport materials and design factories. Derivatives are frequently used to find the maxima and minima of a function. Equations involving derivatives are called differential equations and are fundamental in describing natural phenomena. Derivatives and their generalizations appear in many fields of mathematics, such as complex analysis, functional analysis, differential geometry, measure theory, and abstract algebra.

So, in this course, we will discuss mainly continuity and differentiability of a function of a single variable. We will study the rigorous definition of these two notions, and statements of Rolle's theorem, Lagrange's mean value theorem, and its applications.

1.2 Basic properties of the real numbers

Throughout this course, our focus will be on understanding not only how to solve problems, but also why the fundamental results of calculus are true. This requires a careful study of the real number system, which provides the foundation for the notions of limits, continuity, and differentiability.

Does every bounded sequence converge?

$$(-1)^n$$

Does every bounded increasing sequence converge?

Answer: Yes.

We know the statement is true, but mathematics asks **why it is true**. This naturally motivates the study of the real numbers. Before studying sequences and series, we must understand the number system in which they live. The most important property of the real numbers is **completeness**. Today, we begin with the basic properties of the real numbers.

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c \subset \mathbb{C}$$

The students are expected to be familiar with the algebraic operations on the real numbers, namely addition, additive inverse, multiplication, and multiplicative inverse. They may also wish to briefly revisit the basic notions of sets, relations, binary relations, partially ordered sets, and totally ordered sets. In this course, we shall begin with the fact that $(\mathbb{R}, \leq)$ is a totally ordered set. One of the distinguishing features of the real numbers is that any two real numbers can be compared. The first fundamental property of this order relation is the

Law of Trichotomy

The Law of Trichotomy states that for any two real numbers x and y , exactly one of the following statements holds:

1. $x = y$;
2. $x < y$;
3. $y < x$.

1.2.1 Upper and lower bounds

Definition 1. Let $A \subset \mathbb{R}$ be nonempty. A real number α is called an upper bound of A if for each $x \in A$, we have $x \leq \alpha$.

$$\text{Quantifiers:} \quad \forall x \in A (x \leq \alpha).$$

Geometrically, this means that elements of A are to the left of α on the number line.



Figure 1: α is an upper bound of A

A real number α is not an upper bound of A if there exists at least one $x \in A$ such that $x > \alpha$. In terms of quantifiers

$$\exists x \in A (x > \alpha).$$

Geometrically, this means that we can find an element of A to the right of α on the number line.

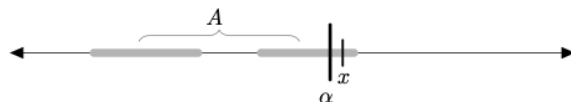


Figure 2: α is not an upper bound of A

If α is an upper bound of A and $\alpha' > \alpha$, then α' is also an upper bound of A .

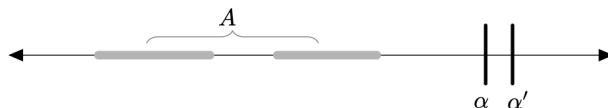


Figure 3: Upper bound of a set is not unique

Exercise 1. Define the notion of a lower bound for a nonempty subset of $\mathbb{R}$.

Exercise 2. Show that $\mathbb{N}$ has a lower bound in $\mathbb{R}$. Does $\mathbb{N}$ have an upper bound in $\mathbb{R}$?

Definition 2. Let $A \subset \mathbb{R}$ be nonempty. We say that A is bounded above in $\mathbb{R}$ if there exists an upper bound of A . Equivalently, there exists $\alpha \in \mathbb{R}$ such that

$$x \leq \alpha, \quad \forall x \in A.$$

In terms of quantifiers, the above definition may be written as

$$\exists \alpha \in \mathbb{R} (\forall x \in A (x \leq \alpha)).$$

Definition 3. Let $A \subset \mathbb{R}$ be nonempty. We say that A is not bounded above in $\mathbb{R}$ if for every $\alpha \in \mathbb{R}$, there exists $x \in A$ (which may depend on α) such that $x > \alpha$.



Figure 4: Set which is not bounded above

In terms of quantifiers, the above definition may be written as

$$\forall \alpha \in \mathbb{R} (\exists x \in A (x > \alpha)).$$

Exercise 3. Using the definition of bounded above, define what it means for a subset of $\mathbb{R}$ to be bounded below. Can you visualize this on the number line? When do you say that a subset $A \subset \mathbb{R}$ is not bounded below in $\mathbb{R}$?

An upper bound of a set need not be an element of the set. For example, if

$$A = (a, b),$$

then b is an upper bound of A , but $b \notin A$. If an upper bound of A is itself an element of A , then it is called the *maximum* of A .

Exercise 4. Suppose that $M, M' \in A$ are both maximum of a nonempty subset $A \subset \mathbb{R}$. Prove that $M = M'$.

Exercise 5. Define the minimum of a nonempty subset $A \subset \mathbb{R}$. Prove that if A has a minimum, then it is unique.

Theorem 1. Let $A \subset \mathbb{R}$ be a nonempty set. If α is a lower bound of A and β is an upper bound of A , then

$$\alpha \leq \beta.$$

Proof. Since A is nonempty, there exists an element $a \in A$. As α is a lower bound of A , we have

$$\alpha \leq a.$$

Similarly, since β is an upper bound of A , we have

$$a \leq \beta.$$

Hence,

$$\alpha \leq a \leq \beta,$$

and therefore

$$\alpha \leq \beta.$$

□

Definition 4. A subset $A \subseteq \mathbb{R}$ is said to be bounded if it is both bounded above and bounded below. Equivalently, there exist real numbers m and M such that $m \leq x \leq M$ for every $x \in A$.

Definition 5. Let $A \subset \mathbb{R}$ be a nonempty set that is bounded above. A real number $\alpha \in \mathbb{R}$ is called the least upper bound (LUB) or supremum of A if

1. α is an upper bound of A ; and
2. if β is any upper bound of A , then $\alpha \leq \beta$.

Notationally we write: $\alpha = \sup A$ or $\alpha = \text{lub } A$.

Definition 6. Let $A \subset \mathbb{R}$ be a nonempty set that is bounded below. A real number $\alpha \in \mathbb{R}$ is called the greatest lower bound (GLB) or infimum of A if

1. α is a lower bound of A ; and
2. if β is any lower bound of A , then $\beta \leq \alpha$.

Notationally we write: $\alpha = \inf A$ or $\alpha = \text{glb } A$.

Example 1. Let

$$A = (0, 1).$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Notice that neither 0 nor 1 belongs to A .

Example 2. Let

$$A = [0, 1].$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

In this case,

$$\max A = 1, \quad \min A = 0.$$

Thus, the supremum and infimum belong to the set.

Example 3. *Let*

$$A = (0, 1].$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Moreover,

$$\max A = 1,$$

but A has no minimum since $0 \notin A$.

Example 4. *Let*

$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Moreover,

$$\max A = 1,$$

but A has no minimum.

Example 5. *Let*

$$A = \mathbb{N}.$$

Then A is bounded below but not bounded above. Here, $\inf A = 1$, whereas $\sup A$ does not exist in $\mathbb{R}$.

Example 6. *Let*

$$A = \{x \in \mathbb{R} : x^2 < 4\}.$$

Since

$$A = (-2, 2),$$

we have

$$\sup A = 2, \quad \inf A = -2.$$

However, A has neither a maximum nor a minimum.

1.2.2 LUB property of the real numbers

Let $A \subset \mathbb{R}$ be nonempty and bounded above. Then there exists $\alpha \in \mathbb{R}$ such that

$$\alpha = \sup A \text{ (or lub } A).$$

$\mathbb{Q}$ does not satisfy the Least Upper Bound Property.

Exercise 6. Construct a nonempty subset of $\mathbb{Q}$ that is bounded above but whose least upper bound does not belong to $\mathbb{Q}$.

Answer: Take $A = \{x \in \mathbb{Q} : x^2 < 2\}$, then $\text{lub } A = \sqrt{2} \notin \mathbb{Q}$.

This example illustrates that the rational numbers have **gaps**, whereas the real numbers do not.

The above result is known as the **Completeness Property** (or **Least Upper Bound Property**) of the real numbers. It is one of the fundamental properties that distinguishes the real number system from the rational number system. In this course, we shall accept this property without proof and use it throughout our study of calculus. A proof of this result requires a rigorous construction of the real numbers, for example, using Dedekind cuts or equivalence classes of Cauchy sequences of rational numbers, which we shall not discuss in this course.

Theorem 2. The set $\mathbb{N}$ is not bounded above in $\mathbb{R}$.

Proof. Suppose, for contradiction, that $\mathbb{N}$ is bounded above in $\mathbb{R}$. Then, by the Completeness Property, the set $\mathbb{N}$ has a least upper bound, say α .

Since $\alpha - 1$ is not an upper bound of $\mathbb{N}$, there exists $n \in \mathbb{N}$ such that

$$n > \alpha - 1.$$

Hence,

$$n + 1 > \alpha.$$

As $n + 1 \in \mathbb{N}$, this contradicts the fact that α is an upper bound of $\mathbb{N}$.

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. □

Theorem 3 (Archimedean Property). For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{N}$ such that

$$nx > y.$$

Proof. Proof by contradiction. Suppose that

$$nx \leq y, \quad \forall n \in \mathbb{N}.$$

Then

$$n \leq \frac{y}{x}, \quad \forall n \in \mathbb{N},$$

since $x > 0$. Thus $\frac{y}{x}$ is an upper bound of $\mathbb{N}$, contradicting the previous theorem. Hence there exists $n \in \mathbb{N}$ such that

$$nx > y.$$

□

Exercise 7. *Prove that the set of integers $\mathbb{Z}$ is neither bounded above nor bounded below in $\mathbb{R}$.*

Recall. For every $x \in \mathbb{R}$, there exists a unique integer $\lfloor x \rfloor$ such that

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

The integer $\lfloor x \rfloor$ is called the *floor* of x . Proving this result using the Archimedean Property is a fun exercise.

Theorem 4 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *Let $a, b \in \mathbb{R}$ be such that $a < b$. Then there exists a rational number $r \in \mathbb{Q}$ satisfying*

$$a < r < b.$$

Proof. Since $b - a > 0$, the Archimedean Property guarantees the existence of $n \in \mathbb{N}$ such that

$$n(b - a) > 1.$$

Let

$$k = \lfloor na \rfloor, \quad m = k + 1.$$

Then

$$k \leq na < k + 1 = m,$$

and hence

$$na < m.$$

We claim that $m < nb$. Suppose, on the contrary, that $m \geq nb$. Then

$$1 = (k + 1) - k = m - k \geq nb - na = n(b - a) > 1,$$

which is impossible. Therefore,

$$m < nb.$$

Consequently,

$$na < m < nb.$$

Dividing throughout by the positive integer n , we obtain

$$a < \frac{m}{n} < b.$$

Since $m, n \in \mathbb{Z}$ and $n > 0$, the number

$$r = \frac{m}{n}$$

is rational. This completes the proof. $\square$

Corollary 1. *Let $a, b \in \mathbb{R}$ be such that $a < b$. Then there exists an irrational number $t \in \mathbb{R} \setminus \mathbb{Q}$ satisfying*

$$a < t < b.$$

Proof. Since

$$a - \sqrt{2} < b - \sqrt{2},$$

the density of $\mathbb{Q}$ in $\mathbb{R}$ guarantees the existence of a rational number r such that

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

Adding $\sqrt{2}$ throughout, we obtain

$$a < r + \sqrt{2} < b.$$

We claim that $r + \sqrt{2}$ is irrational. Indeed, if $r + \sqrt{2}$ were rational, then

$$\sqrt{2} = (r + \sqrt{2}) - r$$

would be the difference of two rational numbers, and hence rational, a contradiction since $\sqrt{2}$ is irrational. Therefore,

$$t = r + \sqrt{2}$$

is an irrational number satisfying

$$a < t < b.$$

This completes the proof. $\square$

Absolute Value. For $x \in \mathbb{R}$, the *absolute value* of x is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

We now recall some useful properties of the absolute value. For all $a, b \in \mathbb{R}$,

1. $|a| \geq 0$ and $|a| = 0$ iff $a = 0$. Moreover, $-|a| \leq a \leq |a|$.
2. Let $\varepsilon > 0$. Then $|a - b| < \varepsilon$ iff $a \in (b - \varepsilon, b + \varepsilon)$.
3. $|ab| = |a||b|$ and $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, provided $b \neq 0$.
4. Triangle Inequality: $|a + b| \leq |a| + |b|$.
5. Reverse Triangle Inequality: $||a| - |b|| \leq |a - b|$.
6. $\max\{a, b\} = \frac{a+b+|a-b|}{2}$ and $\min\{a, b\} = \frac{a+b-|a-b|}{2}$.

2 Sequences

Sequences arise naturally whenever we wish to approximate a quantity. For example, consider the decimal expansion of

$$\frac{1}{7} = 0.142857142857 \dots$$

If we successively retain one more decimal place, we obtain the sequence

$$0.1, 0.14, 0.142, 0.1428, 0.14285, \dots$$

Each term is obtained by keeping one additional decimal place of the decimal expansion of $\frac{1}{7}$, and hence provides a better approximation to $\frac{1}{7}$ than the previous one. For instance, if we require the approximation to be accurate up to 10^{-4} , then any term with at least four decimal places is sufficient.

The notion of a sequence and its convergence provides a rigorous mathematical framework for studying such approximations. Intuitively, a sequence (x_n) converges to a real number x if its terms become arbitrarily close to x as n becomes sufficiently large. In other words, given any prescribed level of accuracy, all terms of the sequence beyond a certain stage approximate x to that accuracy.

2.1 Sequences and their convergence

Definition 7. A sequence of real numbers is a function

$$f : \mathbb{N} \rightarrow \mathbb{R}.$$

For each $n \in \mathbb{N}$, we write

$$x_n = f(n),$$

and call x_n the n -th term of the sequence. The sequence is usually denoted by (x_n) or $\{x_n\}$, and may also be written as

$$(x_n)_n, \quad \{x_n\}_n, \quad \text{or} \quad (x_1, x_2, \dots).$$

Example 7. Some examples of sequences are given below.

1. Fix $c \in \mathbb{R}$ and define $x_n = c$ for all $n \in \mathbb{N}$. Then $(x_n) = (c, c, c, \dots)$. This is called a constant sequence.
2. Let $x_n = \frac{1}{n}$. Then $(x_n) = (1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots)$.
3. Let $x_n = \frac{(-1)^{n+1}}{n}$. Then $(x_n) = (1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, \frac{1}{2k-1}, -\frac{1}{2k}, \dots)$.
4. Let $x_n = (-1)^n$. Then $(x_n) = (-1, 1, -1, 1, \dots)$.
5. Let $x_n = n$. Then $(x_n) = (1, 2, 3, 4, \dots)$.
6. Let $x_n = 2^n$. Then $(x_n) = (2, 4, 8, 16, \dots)$.
7. Let $x_n = \frac{1}{2^n}$. Then $(x_n) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots)$.
8. Let $x_1 = x_2 = 1$ and define $x_n = x_{n-1} + x_{n-2}$ for all $n \geq 3$. Then $(x_n) = (1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$.

Example 8. Let

$$x_1 = 0.1, \quad x_2 = 0.14, \quad x_3 = 0.142, \quad x_4 = 0.1428, \quad \dots,$$

where x_n is obtained by retaining the first n digits after the decimal point in the decimal expansion of

$$\frac{1}{7} = 0.142857142857 \dots$$

Observe that

$$\left| \frac{1}{7} - x_n \right| < \frac{1}{10^n}, \quad \forall n \in \mathbb{N}.$$

Thus, given any prescribed level of accuracy, we can choose n sufficiently large so that x_n approximates $\frac{1}{7}$ to that accuracy. This example motivates the notion of convergence, which we study next.

Definition 8. Let (x_n) be a sequence of real numbers and let $x \in \mathbb{R}$. We say that (x_n) converges to x if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

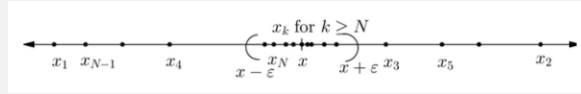
$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

In this case, x is called the limit of the sequence (x_n) . We write

$$x_n \rightarrow x \quad \text{as } n \rightarrow \infty,$$

or equivalently,

$$\lim_{n \rightarrow \infty} x_n = x.$$



In terms of quantifiers: (x_n) converges to x if

$$(\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (\forall n \geq N) (|x - x_n| < \varepsilon).$$

Remark 1. 1. In the definition of convergence, the positive number ε may be chosen arbitrarily small. Nevertheless, for every $\varepsilon > 0$, there always exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

2. Let $\varepsilon > 0$. If (x_n) converges to x , then there exists $N \in \mathbb{N}$ such that

$$x_n \in (x - \varepsilon, x + \varepsilon), \quad \forall n \geq N.$$

Equivalently, the interval $(x - \varepsilon, x + \varepsilon)$ contains all but finitely many terms of the sequence (x_n) , i.e., only finitely many terms of the sequence can stay outside the interval.

Example 9. Consider the constant sequence $x_n = c$ for all $n \in \mathbb{N}$. We claim that

$$\lim_{n \rightarrow \infty} x_n = c.$$

Indeed, let $\varepsilon > 0$ be given. Since

$$|x_n - c| = 0 < \varepsilon, \quad \forall n \in \mathbb{N},$$

we may simply choose $N = 1$. Hence,

$$|x_n - c| < \varepsilon, \quad \forall n \geq N.$$

Therefore, (x_n) converges to c .

Example 10. Consider the sequence $(\frac{1}{n})$. We claim that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Let $\varepsilon > 0$ be given. We have

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ such that

$$N > \frac{1}{\varepsilon}.$$

Hence, for every $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Exercise 8. Using the previous example, show that if $a, b \in \mathbb{R}$ with $a \neq 0$, then the sequence

$$x_n = \frac{1}{an + b}$$

converges. Determine its limit.

Example 11. Consider the sequence

$$a_n = \frac{n^2 + 1}{n^2}.$$

We claim that

$$\lim_{n \rightarrow \infty} a_n = 1.$$

Let $\varepsilon > 0$ be given. We have

$$|a_n - 1| = \left| \frac{n^2 + 1}{n^2} - 1 \right| = \frac{1}{n^2}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ such that

$$N > \frac{1}{\sqrt{\varepsilon}}.$$

Hence, for every $n \geq N$,

$$|a_n - 1| = \frac{1}{n^2} \leq \frac{1}{N^2} < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{n^2 + 1}{n^2} = 1.$$

From the examples above, we see that proving the convergence of a sequence usually has two steps. First, we guess its limit. Then we show that the terms of the sequence get closer and closer to this value by estimating $|x_n - x|$. This raises two natural questions:

How do we guess the limit? & Can a sequence have more than one limit?

The first question becomes easier with practice. We now answer the second question.

Theorem 5 (Uniqueness of the Limit). *If*

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{and} \quad \lim_{n \rightarrow \infty} x_n = y,$$

then $x = y$.

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n \geq N_1.$$

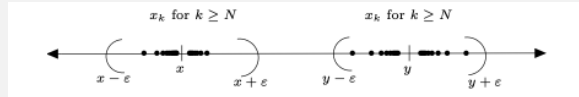
Similarly, since $x_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that

$$|x_n - y| < \frac{\varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$, the triangle inequality gives

$$|x - y| = |x - x_n + x_n - y| \leq |x - x_n| + |x_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $x = y$.



□

Example 12. Consider the sequence $x_n = n$. We claim that this sequence does not converge.

Assume, on the contrary, that (x_n) converges to some $x \in \mathbb{R}$. By Remark 1, following the definition of convergence, the interval

$$(x - 1, x + 1)$$

must contain all but finitely many terms of the sequence. Hence, there exists $N \in \mathbb{N}$ such that

$$n = x_n \in (x - 1, x + 1), \quad \forall n \geq N.$$

In particular,

$$n < x + 1, \quad \forall n \geq N.$$

Thus, the set $\mathbb{N}$ is bounded above by $\max\{N, x + 1\}$, contradicting the fact that $\mathbb{N}$ is not bounded above. Therefore, (x_n) does not converge.

Example 13. Consider the sequence $x_n = (-1)^n$. Observe that the intervals

$$\left(-\frac{3}{2}, -\frac{1}{2}\right) \quad \text{and} \quad \left(\frac{1}{2}, \frac{3}{2}\right)$$

each contain infinitely many terms of the sequence. Hence, no interval containing a possible limit can contain all but finitely many terms of the sequence. Therefore, $((-1)^n)$ is not convergent.

Converge

$$\lim_{n \rightarrow \infty} x_n = x \text{ if } (\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (\forall n \geq N) (|x - x_n| < \varepsilon).$$

Diverge

$$(\forall x \in \mathbb{R}) (\exists \varepsilon > 0) (\forall N \in \mathbb{N}) (\exists n \geq N) (|x_n - x| \geq \varepsilon).$$

A sequence that is not convergent is called *divergent*. This does *not* mean that the terms of the sequence must tend to infinity. A divergent sequence may oscillate, as in the sequence $((-1)^n)$, or it may become arbitrarily large, as in the sequence (n) . Thus, there are different ways in which a sequence can fail to converge. To prove that a sequence is convergent, we must identify its limit and verify the definition of convergence. On the other hand, to prove that a sequence is divergent, it is enough to show that no real number can be its limit.

2.2 Properties of convergent sequences

Definition 9. A sequence (x_n) of real numbers is said to be bounded if there exists a constant $C > 0$ such that

$$|x_n| \leq C, \quad \forall n \in \mathbb{N}.$$

Example 14. The following are some examples of bounded and unbounded sequences.

1. Every constant sequence is bounded.
2. The sequence $\left(\frac{1}{n}\right)$ is bounded.
3. The sequence $((-1)^n)$ is bounded.
4. The sequence $\left(\frac{n^2+1}{n^2}\right)$ is bounded.
5. The sequence $(\sin n)$ is bounded.

Definition 10. A sequence (x_n) of real numbers is said to be unbounded if, for every $C > 0$, there exists $n \in \mathbb{N}$ such that $|x_n| > C$.

Example 15. The sequences

$$(n^2), \quad (n^n), \quad (n)$$

are unbounded.

Theorem 6. Every convergent sequence of real numbers is bounded.

Proof. Let (x_n) be a sequence converging to $x \in \mathbb{R}$. Choose $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < 1, \quad \forall n \geq N.$$

By the triangle inequality,

$$|x_n| = |x_n - x + x| \leq |x_n - x| + |x| < 1 + |x|, \quad \forall n \geq N.$$

Now let

$$C = \max\{|x_1|, \dots, |x_{N-1}|, 1 + |x|\}.$$

Then

$$|x_n| \leq C, \quad \forall n \in \mathbb{N}.$$

Hence, (x_n) is bounded. □

The converse of the above result is not true. Can you recall an example?

Theorem 7. Let (x_n) be a sequence of real numbers such that

$$\lim_{n \rightarrow \infty} x_n = x,$$

where $x \neq 0$. Then there exists $N \in \mathbb{N}$ such that

$$|x_n| > \frac{|x|}{2}, \quad \forall n \geq N.$$

Proof. Choose

$$\varepsilon = \frac{|x|}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{|x|}{2}, \quad \forall n \geq N.$$

By the triangle inequality,

$$|x| = |x - x_n + x_n| \leq |x - x_n| + |x_n| < \frac{|x|}{2} + |x_n|.$$

Hence,

$$|x_n| > \frac{|x|}{2}, \quad \forall n \geq N.$$

This completes the proof. □

Theorem 8. Let (a_n) be a sequence of real numbers such that

$$\lim_{n \rightarrow \infty} a_n = a.$$

If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.

Proof. Suppose, on the contrary, that $a < 0$. Then there exists $\delta > 0$ such that

$$a + \delta < 0.$$

Since $a_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that

$$a_n \in (a - \delta, a + \delta), \quad \forall n \geq N.$$

Since $a + \delta < 0$, every point in the interval $(a - \delta, a + \delta)$ is negative. Thus,

$$a_n < 0, \quad \forall n \geq N,$$

which contradicts the assumption that $a_n \geq 0$ for all $n \in \mathbb{N}$. Therefore, $a \geq 0$. □

Exercise 9. Let (u_n) and (v_n) be two convergent sequences such that

$$u_n \leq v_n, \quad \forall n \in \mathbb{N}.$$

Show that

$$\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n.$$

Given two sequences (x_n) and (y_n) , we can construct new sequences by adding, multiplying, or multiplying by a real number. A natural question is: if the original sequences are convergent, what can we say about these newly constructed sequences? In this section, we show that convergence is preserved under these basic algebraic operations.

Theorem 9 (Algebra of Convergent Sequences). Let (x_n) and (y_n) be sequences of real numbers such that $\lim_{n \rightarrow \infty} x_n = x$, $\lim_{n \rightarrow \infty} y_n = y$, and let $\alpha \in \mathbb{R}$. Then:

1. $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$.
2. $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x$.
3. $\lim_{n \rightarrow \infty} (x_n y_n) = xy$.
4. If $x_n > 0$ for all $n \in \mathbb{N}$ and $x > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}$.

Proof. Let $\varepsilon > 0$ be given.

(1) Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n \geq N_1.$$

Similarly, since $y_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that

$$|y_n - y| < \frac{\varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence,

$$\lim_{n \rightarrow \infty} (x_n + y_n) = x + y.$$

(2) If $\alpha = 0$, then the result is obvious. Assume $\alpha \neq 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{|\alpha|}, \quad \forall n \geq N.$$

Hence,

$$|\alpha x_n - \alpha x| = |\alpha||x_n - x| < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x.$$

(3) Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n| - |x| \leq ||x_n| - |x|| \leq |x_n - x| < 1, \quad \forall n \geq N_1.$$

Also, there exist $N_2, N_3 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2(1 + |y|)}$$

for all $n \geq N_2$, and

$$|y_n - y| < \frac{\varepsilon}{2(1 + |x|)}$$

for all $n \geq N_3$. Let $N = \max\{N_1, N_2, N_3\}$. Then

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x_n y + x_n y - xy| \\ &\leq |x_n||y_n - y| + |y||x_n - x| \\ &\leq (|x| + 1) \frac{\varepsilon}{2(1 + |x|)} + |y| \frac{\varepsilon}{2(1 + |y|)} < \varepsilon. \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} (x_n y_n) = xy.$$

(4) Since $x > 0$, by the previous Theorem 7 there exists $N_1 \in \mathbb{N}$ such that

$$x_n > \frac{x}{2}, \quad \forall n \geq N_1.$$

Also, there exists $N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{x^2 \varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\left| \frac{1}{x_n} - \frac{1}{x} \right| = \frac{|x_n - x|}{x x_n} \leq \frac{2}{x^2} |x_n - x| < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}.$$

□

Exercise 10. In Part (4) of the above theorem, the assumption $x > 0$ is essential. What can go wrong if $x = 0$? Give an example to show that the conclusion need not hold.

2.3 Cauchy sequences

So far, every time we proved that a sequence converges, we first guessed its limit and then verified the definition of convergence. This naturally raises the following question:

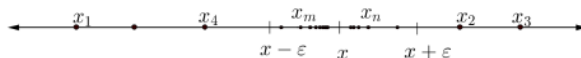
Can we decide whether a sequence is convergent without knowing its limit?

Suppose that (x_n) converges to x . Let $\varepsilon > 0$ be given. Then there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

Hence, for any $m, n \geq N$, the triangle inequality gives

$$|x_n - x_m| \leq |x_n - x| + |x_m - x| < \varepsilon + \varepsilon = 2\varepsilon.$$



Thus, when the indices are sufficiently large, the terms of a convergent sequence become arbitrarily close to each other. This observation motivates the following definition.

Definition 11 (Cauchy Sequence). A sequence (x_n) of real numbers is said to be Cauchy if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x_m| < \varepsilon, \quad \forall m, n \geq N.$$

Example 16. Every convergent sequence is a Cauchy sequence. Is the converse true?

Theorem 10. [Cauchy Criterion] A sequence of real numbers is Cauchy if and only if it is convergent.

Proof. We have already seen that every convergent sequence is Cauchy. We now prove the converse.

Let $E = \{x \in \mathbb{R} : \exists N \in \mathbb{N} \text{ such that } x < x_n, \forall n \geq N\}$.

Let $\delta > 0$. Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|x_m - x_n| < \delta$ for all

$m, n \geq N_0$. In particular, $|x_{N_0} - x_n| < \delta$ for all $n \geq N_0$. Hence, $x_n \in (x_{N_0} - \delta, x_{N_0} + \delta)$ for all $n \geq N_0$. In particular, $x_n < x_{N_0} + \delta$ for all $n \geq N_0$.

Claim 1. $x_{N_0} - \delta \in E$.

Indeed, taking $N = N_0$, we have $x_n > x_{N_0} - \delta$ for all $n \geq N_0$.

Claim 2. $x_{N_0} + \delta$ is an upper bound of E .

Suppose not. Then there exists $x \in E$ such that $x > x_{N_0} + \delta$. Since $x \in E$, there exists $N \in \mathbb{N}$ such that $x_n > x$ for all $n \geq N$. Hence, for all $n \geq \max\{N, N_0\}$,

$$x_n > x > x_{N_0} + \delta,$$

which contradicts the inequality $x_n < x_{N_0} + \delta$ for all $n \geq N_0$.

Thus, E is nonempty and bounded above. By the completeness of $\mathbb{R}$, let $\ell = \sup E$.

Claim 3. $x_n \rightarrow \ell$.

Let $\varepsilon > 0$. Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|x_n - x_{N_0}| < \varepsilon/2$ for all $n \geq N_0$.

By Claim 1, $x_{N_0} - \varepsilon/2 \in E$, and hence $x_{N_0} - \varepsilon/2 \leq \ell$. By Claim 2, $\ell \leq x_{N_0} + \varepsilon/2$. Therefore, $|x_{N_0} - \ell| \leq \varepsilon/2$.

Now, for every $n \geq N_0$,

$$|x_n - \ell| \leq |x_n - x_{N_0}| + |x_{N_0} - \ell| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $x_n \rightarrow \ell$. □

Example 17. The sequence $(\frac{1}{n})$ is a Cauchy sequence.

Proof. Let $\varepsilon > 0$ be given. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then, for any $m, n \geq N$, without loss of generality, assume $m \leq n$. Hence,

$$\left| \frac{1}{m} - \frac{1}{n} \right| = \frac{1}{m} - \frac{1}{n} < \frac{1}{m} \leq \frac{1}{N} < \varepsilon.$$

Therefore, $(\frac{1}{n})$ is a Cauchy sequence. □

Example 18. The sequence $((-1)^n)$ is not a Cauchy sequence.

Proof. Choose $\varepsilon = 1$. Let $N \in \mathbb{N}$ be arbitrary. Take

$$m = \begin{cases} N + 1, & \text{if } N \text{ is even,} \\ N + 2, & \text{if } N \text{ is odd,} \end{cases} \quad \text{and} \quad n = m + 1.$$

Then $m, n \geq N$, $(-1)^m = 1$, and $(-1)^n = -1$. Hence,

$$|(-1)^m - (-1)^n| = |1 - (-1)| = 2 > 1 = \varepsilon.$$

Therefore, $((-1)^n)$ is not a Cauchy sequence. □

Example 19. *Let*

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \quad n \in \mathbb{N}.$$

We show, using the Cauchy Criterion, that (s_n) is not convergent.

Proof. By the Cauchy Criterion, it is enough to show that (s_n) is not a Cauchy sequence.

Choose $\varepsilon = \frac{1}{2}$ and let $N \in \mathbb{N}$ be arbitrary. Take $n \geq N$ and let $m = 2n$. Then

$$\begin{aligned} |s_m - s_n| &= |s_{2n} - s_n| \\ &= \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \\ &\geq \underbrace{\frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ terms}} = \frac{1}{2}. \end{aligned}$$

Thus, for every $N \in \mathbb{N}$, there exist $m, n \geq N$ such that

$$|s_m - s_n| \geq \frac{1}{2}.$$

Hence, (s_n) is not a Cauchy sequence. Therefore, by the Cauchy Criterion, (s_n) is not convergent. $\square$

2.4 Monotone sequences

Definition 12. *A sequence (x_n) is said to be increasing if*

$$x_n \leq x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 13. *A sequence (x_n) is said to be strictly increasing if*

$$x_n < x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 14. *A sequence (x_n) is said to be decreasing if*

$$x_n \geq x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 15. *A sequence (x_n) is said to be strictly decreasing if*

$$x_n > x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 16. A sequence is said to be monotone if it is either increasing or decreasing.

Remark 2. Every increasing sequence is bounded below by x_1 . Hence, an increasing sequence is bounded if and only if it is bounded above. Similarly, every decreasing sequence is bounded above by x_1 . Therefore, a decreasing sequence is bounded if and only if it is bounded below.

Example 20. The following are examples of monotone sequences.

1. The constant sequence (c) is both increasing and decreasing.
2. $(\frac{1}{n})$ is strictly decreasing.
3. (n) is strictly increasing.
4. (2^n) is strictly increasing.
5. $(\frac{n^2+1}{n^2}) = (1 + \frac{1}{n^2})$ is strictly decreasing.
6. $((-1)^n)$ is not monotone.

Theorem 11 (Monotone Convergence Theorem). An increasing sequence of real numbers is convergent if and only if it is bounded above.

Proof. Let (x_n) be an increasing sequence that is bounded above. We show that (x_n) is convergent.

Let

$$E = \{x_n : n \in \mathbb{N}\}.$$

Since E is nonempty and bounded above, by the completeness property of $\mathbb{R}$, the set E has a least upper bound. Let

$$\ell = \sup E.$$

Let $\varepsilon > 0$ be given. Since $\ell - \varepsilon$ is not an upper bound of E , there exists $N \in \mathbb{N}$ such that

$$x_N > \ell - \varepsilon.$$

As (x_n) is increasing, for every $n \geq N$,

$$\ell - \varepsilon < x_N \leq x_n \leq \ell < \ell + \varepsilon.$$

Hence,

$$|x_n - \ell| < \varepsilon, \quad \forall n \geq N.$$

Therefore,

$$\lim_{n \rightarrow \infty} x_n = \ell.$$

The converse follows immediately from the fact that every convergent sequence is bounded. $\square$

Exercise 11. Show that a decreasing sequence of real numbers is convergent if and only if it is bounded below.

Notation. The notation

$$x_n \downarrow x$$

means that the sequence (x_n) is decreasing and converges to x . Similarly, the notation

$$x_n \uparrow x$$

means that the sequence (x_n) is increasing and converges to x .

Example 21. Let $0 \leq r < 1$. We show that $\lim_{n \rightarrow \infty} r^n = 0$.

If $r = 0$, then there is nothing to prove. So assume that $0 < r < 1$. Since $r^{n+1} < r^n$ for every $n \in \mathbb{N}$, the sequence (r^n) is decreasing. Moreover, it is bounded below by 0. Hence, by the Monotone Convergence Theorem, (r^n) converges. Let $\lim_{n \rightarrow \infty} r^n = \ell$.

By the Algebra of Convergent Sequences, $\lim_{n \rightarrow \infty} r^{n+1} = r\ell$. On the other hand, r^{n+1} is just the shifted sequence of (r^n) , so $\lim_{n \rightarrow \infty} r^{n+1} = \ell$. By the Uniqueness of Limits, $r\ell = \ell$. Since $0 \leq r < 1$, we have $(1 - r)\ell = 0$, and therefore $\ell = 0$. Hence,

$$\lim_{n \rightarrow \infty} r^n = 0.$$

Example 22. Let $0 \leq r < 1$, and define $s_n := 1 + r + r^2 + \cdots + r^n$.

If $r = 0$, then (s_n) is the constant sequence 1, and hence it is convergent. So assume that $0 < r < 1$. Since $s_{n+1} = s_n + r^{n+1} > s_n$, the sequence (s_n) is increasing.

Moreover, $s_n = \frac{1 - r^{n+1}}{1 - r} \leq \frac{1}{1 - r}$, so (s_n) is bounded above. Hence, by the Monotone Convergence Theorem, (s_n) converges.

Finally, $s_n = \frac{1 - r^{n+1}}{1 - r} = \frac{1}{1 - r} - \frac{r^{n+1}}{1 - r}$. Since $\lim_{n \rightarrow \infty} r^{n+1} = 0$, it follows from the Algebra of Convergent Sequences that

$$\lim_{n \rightarrow \infty} s_n = \frac{1}{1 - r}.$$

2.5 Sandwich theorem

In many situations, we may be able to guess the limit of a sequence but find it difficult to prove the convergence directly. A useful idea is to compare the given sequence with two simpler sequences whose limits are already known. If the given sequence always lies between these two sequences, then it is forced to have the same limit. This idea is known as the *Sandwich Theorem*.

Theorem 12 (Sandwich Theorem). *Let (x_n) , (y_n) and (z_n) be sequences of real numbers such that*

1. $\lim_{n \rightarrow \infty} x_n = \alpha$ and $\lim_{n \rightarrow \infty} y_n = \alpha$;
2. $x_n \leq z_n \leq y_n$ for all $n \in \mathbb{N}$.

Then

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow \alpha$, there exists $N_1 \in \mathbb{N}$ such that $x_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$ for all $n \geq N_1$. Similarly, there exists $N_2 \in \mathbb{N}$ such that $y_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$ for all $n \geq N_2$.

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\alpha - \varepsilon < x_n \leq z_n \leq y_n < \alpha + \varepsilon.$$

Hence,

$$|z_n - \alpha| < \varepsilon, \quad \forall n \geq N.$$

Therefore,

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

□

Example 23. *Define the sequence (z_n) by*

$$z_n = \frac{\sin n}{n}, \quad n \in \mathbb{N}.$$

We show that $\lim_{n \rightarrow \infty} z_n = 0$.

Since

$$-1 \leq \sin n \leq 1,$$

we have

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

We already know that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \left(-\frac{1}{n}\right) = 0.$$

Hence, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0.$$

Example 24. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} \right) = 1.$$

Proof. Let $a_n := \sum_{k=1}^n \frac{1}{\sqrt{n^2+k}}$. Since $\frac{1}{\sqrt{n^2+n}} \leq \frac{1}{\sqrt{n^2+k}} \leq \frac{1}{\sqrt{n^2+1}}$, for every $1 \leq k \leq n$, we obtain $u_n \leq a_n \leq v_n$, where $u_n := \frac{n}{\sqrt{n^2+n}}$ and $v_n := \frac{n}{\sqrt{n^2+1}}$.

Now, $u_n = \frac{1}{\sqrt{1+\frac{1}{n}}} \rightarrow 1$, and $v_n = \frac{1}{\sqrt{1+\frac{1}{n^2}}} \rightarrow 1$. Hence, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} a_n = 1.$$

□

Example 25. Let $a > 0$. We show that

$$\lim_{n \rightarrow \infty} a^{1/n} = 1.$$

Proof. If $a = 1$, there is nothing to prove. Assume first that $a > 1$. Write $a^{1/n} = 1 + x_n$, where $x_n > 0$. Then $a = (1 + x_n)^n$. By Bernoulli's inequality, $(1 + x_n)^n \geq 1 + nx_n$. Hence, $a \geq 1 + nx_n$, which implies $0 < x_n \leq \frac{a-1}{n}$.

Let $\varepsilon > 0$ be given. By the Archimedean Property, choose $N \in \mathbb{N}$ such that $\frac{a-1}{N} < \varepsilon$.

Then, for every $n \geq N$, $0 < x_n \leq \frac{a-1}{n} \leq \frac{a-1}{N} < \varepsilon$. Therefore, $|a^{1/n} - 1| = x_n < \varepsilon$, and hence $a^{1/n} \rightarrow 1$.

Now suppose that $0 < a < 1$. Then $\frac{1}{a} > 1$, and by the first part, $\left(\frac{1}{a}\right)^{1/n} \rightarrow 1$. Since $a^{1/n} = \frac{1}{(1/a)^{1/n}}$, the Algebra of Convergent Sequences gives $a^{1/n} \rightarrow 1$. □

Example 26. Let $0 < a < b$. We show that

$$\lim_{n \rightarrow \infty} (a^n + b^n)^{1/n} = b.$$

Proof. Since $a^n > 0$, we have $b^n \leq a^n + b^n \leq 2b^n$. Taking the n -th root, we obtain $b \leq (a^n + b^n)^{1/n} \leq 2^{1/n}b$.

We already know that $\lim_{n \rightarrow \infty} 2^{1/n} = 1$. Hence, $\lim_{n \rightarrow \infty} 2^{1/n}b = b$. Therefore, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} (a^n + b^n)^{1/n} = b.$$

□

Exercise 12. Let (a_n) be a bounded sequence of real numbers and let (x_n) be a sequence such that $x_n \rightarrow 0$.

1. Show that $\lim_{n \rightarrow \infty} a_n x_n = 0$.

2. Deduce that

$$\lim_{n \rightarrow \infty} r^n = 0, \quad \text{for every } |r| < 1.$$

(Hint. Write $r^n = |r|^n \left(\frac{r}{|r|}\right)^n$ when $r \neq 0$.)

Exercise 13 (Some important limits). Solve the following exercises.

1. Show that $\lim_{n \rightarrow \infty} n^{1/n} = 1$.

2. Let $a > 1$. Show that $\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0$.

3. Let $a \in \mathbb{R}$. Show that $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$.

4. Let (x_n) be a sequence converging to 0. Define $s_n := \frac{x_1 + x_2 + \cdots + x_n}{n}$. Show that (s_n) also converges to 0.

5. Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Show that $\frac{x_1 y_n + x_2 y_{n-1} + \cdots + x_n y_1}{n} \rightarrow xy$.

2.6 Sequences diverging to $\pm\infty$

Definition 17. Let (x_n) be a real sequence. We say that (x_n) diverges to $+\infty$ (or simply diverges to ∞) if, for every $R \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies x_n > R.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = +\infty.$$

Definition 18. Let (x_n) be a real sequence. We say that (x_n) diverges to $-\infty$ if, for every $R \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies x_n < R.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = -\infty.$$

Example 27. The sequence (n^2) diverges to $+\infty$, whereas the sequence $(-n)$ diverges to $-\infty$. That is,

$$\lim_{n \rightarrow \infty} n^2 = +\infty \quad \text{and} \quad \lim_{n \rightarrow \infty} (-n) = -\infty.$$

Example 28. Define a sequence (x_n) by

$$x_{2k-1} = 1, \quad x_{2k} = 2k, \quad k \in \mathbb{N}.$$

The sequence (x_n) is unbounded but does not diverge to $+\infty$.

Indeed, let $R = 1$. For any $N \in \mathbb{N}$, choose $n = 2N - 1$. Then $n \geq N$, but

$$x_n = x_{2N-1} = 1 \not\geq R.$$

Hence (x_n) does not diverge to $+\infty$.

Example 29. Let $a > 1$. Then

$$\lim_{n \rightarrow \infty} a^n = +\infty.$$

Indeed, write $a = 1 + h$ for some $h > 0$. By Bernoulli's inequality,

$$a^n = (1 + h)^n \geq 1 + nh > nh.$$

Let $R \in \mathbb{R}$ be given. By the Archimedean property, there exists $N \in \mathbb{N}$ such that

$$Nh > R.$$

Then, for every $n \geq N$,

$$a^n > nh \geq Nh > R.$$

Hence $a^n \rightarrow +\infty$.

Example 30. The sequence $((n!)^{1/n})$ diverges to $+\infty$.

Let $R > 0$ be given. Since

$$\lim_{n \rightarrow \infty} \frac{R^n}{n!} = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\frac{R^n}{n!} < 1, \quad n \geq N.$$

Thus

$$R^n < n!,$$

and therefore

$$R < (n!)^{1/n}, \quad n \geq N.$$

Hence

$$(n!)^{1/n} \longrightarrow +\infty.$$

Example 31. Consider the sequence

$$x_n = (-1)^n.$$

This sequence is divergent. Moreover, it diverges neither to $+\infty$ nor to $-\infty$. Indeed, the sequence is bounded since $-1 \leq x_n \leq 1$ for all n . Hence, it cannot diverge to $+\infty$ or to $-\infty$. On the other hand, we have already shown that (x_n) is not convergent. Therefore, (x_n) is divergent, but not divergent to $\pm\infty$.

Example 32. Define

$$x_n := \sum_{k=1}^n \frac{1}{k}, \quad n \in \mathbb{N}.$$

We show that (x_n) diverges to $+\infty$.

Let $R \in \mathbb{R}$ be given. By the Archimedean Property, choose $m \in \mathbb{N}$ such that

$$\frac{m}{2} > R.$$

Now let $N := 2^m$. For every $n \geq N$, we have

$$x_n \geq x_{2^m}.$$

Moreover,

$$\begin{aligned} x_{2^m} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots \\ &\quad + \left(\frac{1}{2^{m-1}+1} + \cdots + \frac{1}{2^m}\right). \end{aligned}$$

Each group after the first two contains at least

$$2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2},$$

where $j = 1, 2, \dots, m$. Hence

$$x_{2^m} \geq 1 + \frac{m}{2} > R.$$

Therefore,

$$x_n \geq x_{2^m} > R, \quad n \geq N.$$

This proves that

$$\lim_{n \rightarrow \infty} x_n = +\infty.$$

Example 33. Consider the sequence

$$a_n := \left(1 + \frac{1}{n}\right)^n, \quad n \in \mathbb{N}.$$

We show that (a_n) is increasing and bounded above. Consequently, it is convergent.

Step 1. (a_n) is increasing.

Consider the $(n+1)$ numbers

$$\underbrace{\frac{n+1}{n}, \frac{n+1}{n}, \dots, \frac{n+1}{n}}_{n \text{ times}}, 1.$$

By the Arithmetic Mean–Geometric Mean inequality,

$$\frac{n \left(\frac{n+1}{n} \right) + 1}{n+1} \geq \left[\left(\frac{n+1}{n} \right)^n \right]^{\frac{1}{n+1}}.$$

Since the arithmetic mean is

$$\frac{(n+1) + 1}{n+1} = 1 + \frac{1}{n+1},$$

we obtain

$$1 + \frac{1}{n+1} \geq \left(1 + \frac{1}{n}\right)^{\frac{n}{n+1}}.$$

Raising both sides to the power $n+1$ gives

$$\left(1 + \frac{1}{n+1}\right)^{n+1} \geq \left(1 + \frac{1}{n}\right)^n.$$

Hence (a_n) is increasing.

Step 2. (a_n) is bounded above.

Using the Binomial Theorem,

$$\begin{aligned} a_n &= \left(1 + \frac{1}{n}\right)^n \\ &= 1 + 1 + \frac{n(n-1)}{2!n^2} + \frac{n(n-1)(n-2)}{3!n^3} + \dots + \frac{1}{n^n} \\ &< 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(n-1)!} + \frac{1}{n!} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \\ &= 3. \end{aligned}$$

Thus (a_n) is bounded above. Also, we used the fact that $k! \geq 2^{k-1}$ for every $k \geq 1$, and so we have

$$\frac{1}{k!} \leq \frac{1}{2^{k-1}}.$$

Therefore, by the Monotone Convergence Theorem, (a_n) is convergent.

We denote its limit by

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

The number e is one of the most important constants in mathematics.

Exercise 14. Let

$$x_n := 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

1. Show that (x_n) is monotonically increasing.
2. Show that (x_n) is bounded above.
3. Deduce that (x_n) is convergent.
4. Show that

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

2.7 Ratio and root tests

Theorem 13 (Ratio Test). Let (u_n) be a sequence of positive real numbers such that

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \ell.$$

Then the following holds.

1. If $0 \leq \ell < 1$, then

$$\lim_{n \rightarrow \infty} u_n = 0.$$

2. If $\ell > 1$, then

$$\lim_{n \rightarrow \infty} u_n = +\infty.$$

Remark 3. If $\ell = 1$, no conclusion can be drawn.

For example,

$$u_n = n$$

satisfies

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1,$$

but (u_n) diverges to $+\infty$.

On the other hand,

$$u_n = \frac{n+1}{n}$$

also satisfies

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1,$$

but

$$\lim_{n \rightarrow \infty} u_n = 1.$$

Theorem 14 (Root Test). *Let (u_n) be a sequence of positive real numbers such that*

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \ell.$$

Then the following holds.

1. *If $0 \leq \ell < 1$, then*

$$\lim_{n \rightarrow \infty} u_n = 0.$$

2. *If $\ell > 1$, then*

$$\lim_{n \rightarrow \infty} u_n = +\infty.$$

Remark 4. *If $\ell = 1$, no conclusion can be drawn.*

For example,

$$u_n = n$$

satisfies

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = 1,$$

but (u_n) diverges to $+\infty$.

On the other hand,

$$u_n = \frac{1}{n}$$

also satisfies

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = 1,$$

but

$$\lim_{n \rightarrow \infty} u_n = 0.$$

Example 34. Show that

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0.$$

Indeed,

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2^{n+1}/(n+1)!}{2^n/n!} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0.$$

Hence, the conclusion follows from the Ratio Test.

Example 35. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0.$$

Indeed,

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{1}{2}\right)^n} = \frac{1}{2} < 1.$$

Hence, the conclusion follows from the Root Test.

2.8 Subsequences

Definition 19. Let (r_n) be a strictly increasing sequence of natural numbers. If (x_n) is any sequence, then the sequence

$$(x_{r_n})$$

is called a subsequence of (x_n) .

One can view a subsequence of (x_n) as an infinite part of the sequence obtained by deleting some terms, while keeping the remaining terms in the same order.

Example 36. Let

$$x_n = 2^n, \quad n \in \mathbb{N}.$$

Then

$$\{2^{2n}\}_{n \in \mathbb{N}} = \{4, 16, 64, \dots\}$$

is a subsequence of (x_n) .

Here, the corresponding indices are

$$r_n = 2n, \quad n \in \mathbb{N},$$

which form a strictly increasing sequence.

Remark 5. A subsequence of (x_n) can also be written as $(x_{f(n)})$, where

$$f : \mathbb{N} \rightarrow \mathbb{N}$$

is a strictly increasing function. Here, the n th term of the subsequence is the $f(n)$ th term of the original sequence.

Example 37. Let (x_n) be any sequence.

1. The sequence

$$(x_2, x_4, x_6, \dots)$$

is called the even subsequence of (x_n) . Here,

$$r_n = 2n, \quad n \in \mathbb{N}.$$

2. The sequence

$$(x_1, x_3, x_5, \dots)$$

is called the odd subsequence of (x_n) . Here,

$$r_n = 2n - 1, \quad n \in \mathbb{N}.$$

Theorem 15. Let (x_n) be a sequence converging to x . If (x_{r_n}) is a subsequence of (x_n) , then

$$x_{r_n} \rightarrow x.$$

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

Since (r_n) is a strictly increasing sequence of natural numbers, we have $r_n \geq n$ for every $n \in \mathbb{N}$. Hence, if $n \geq N$, then

$$r_n \geq n \geq N,$$

and therefore

$$|x_{r_n} - x| < \varepsilon.$$

Thus,

$$x_{r_n} \rightarrow x.$$

□

Exercise 15. Let (x_n) be a sequence. Suppose that both the even subsequence (x_{2n}) and the odd subsequence (x_{2n-1}) converge to the same limit x . Show that the original

sequence (x_n) also converges to x .

Example 38. Show that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2n}\right)^n = \sqrt{e}.$$

Solution. Let

$$x_n = \left(1 + \frac{1}{n}\right)^n.$$

We have already shown that $x_n \rightarrow e$. Hence, every subsequence of (x_n) also converges to e . In particular,

$$x_{2n} = \left(1 + \frac{1}{2n}\right)^{2n} \rightarrow e.$$

Therefore,

$$\left[\left(1 + \frac{1}{2n}\right)^n\right]^2 \rightarrow e.$$

Since $\left(1 + \frac{1}{2n}\right)^n > 0$ for every n , it follows that

$$\left(1 + \frac{1}{2n}\right)^n \rightarrow \sqrt{e}.$$

Theorem 16 (Existence of a Monotone Subsequence). *Every real sequence has a monotone subsequence.*

Theorem 17 (Bolzano–Weierstrass Theorem). *Every bounded real sequence has a convergent subsequence.*

The proofs of the above two theorems are beyond the scope of this course and are therefore omitted. Interested students may consult a standard text on Real Analysis for their proofs.

Example 39. Assume that the sequence $(n^{1/n})$ is convergent. Show that

$$\lim_{n \rightarrow \infty} n^{1/n} = 1.$$

Solution. Let

$$n^{1/n} \rightarrow \ell.$$

Since $((2n)^{1/(2n)})$ is a subsequence of $(n^{1/n})$, we also have

$$(2n)^{1/(2n)} \rightarrow \ell.$$

Hence

$$\left((2n)^{1/(2n)}\right)^2 \longrightarrow \ell^2.$$

On the other hand,

$$\left((2n)^{1/(2n)}\right)^2 = 2^{1/n} n^{1/n}.$$

Since

$$2^{1/n} \longrightarrow 1,$$

it follows from the algebra of limits that

$$2^{1/n} n^{1/n} \longrightarrow \ell.$$

Therefore,

$$\ell^2 = \ell.$$

Since $n^{1/n} \geq 1$ for every n , we have $\ell \geq 1$. Thus,

$$\ell = 1.$$

Example 40. Show that the sequence $((-1)^n)$ is divergent.

Solution. Suppose, on the contrary, that $((-1)^n)$ converges to some real number ℓ . Then every subsequence must also converge to ℓ .

However, the even subsequence is

$$((-1)^{2n}) = (1, 1, 1, \dots),$$

which converges to 1, whereas the odd subsequence is

$$((-1)^{2n-1}) = (-1, -1, -1, \dots),$$

which converges to -1 .

Since a convergent sequence cannot have two subsequences with different limits, we obtain a contradiction. Hence

$$((-1)^n)$$

is divergent.

Exercise 16. Apply the Bolzano–Weierstrass Theorem to solve the following problems:

1. Show that the sequence $(\sin n)$ has a convergent subsequence.
2. Prove that every bounded sequence has a monotone convergent subsequence.
3. True or False: A sequence (x_n) is bounded if and only if every subsequence of (x_n) has a convergent subsequence.